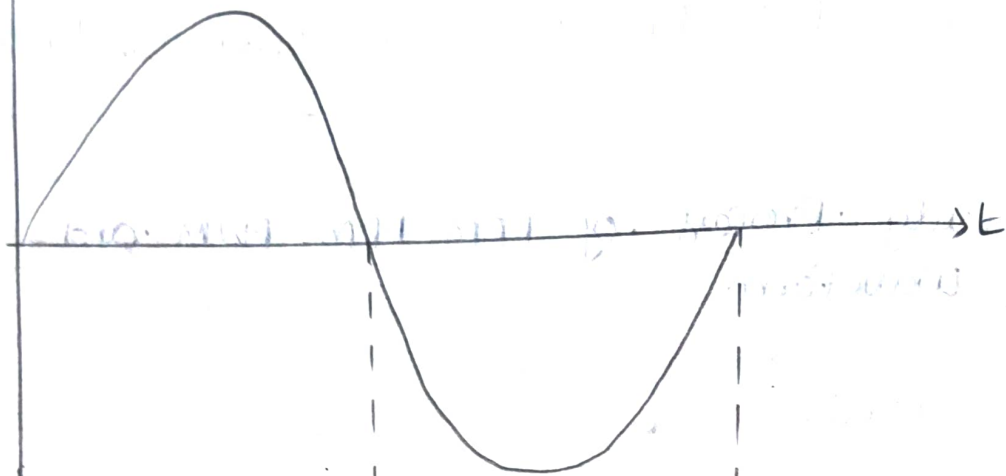
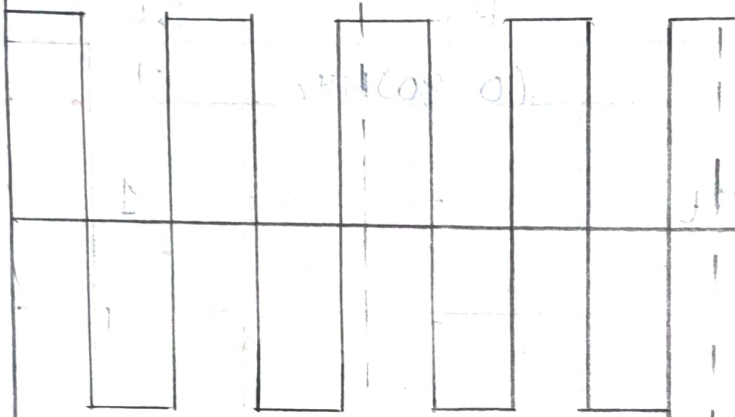


MODEL GRAPH:

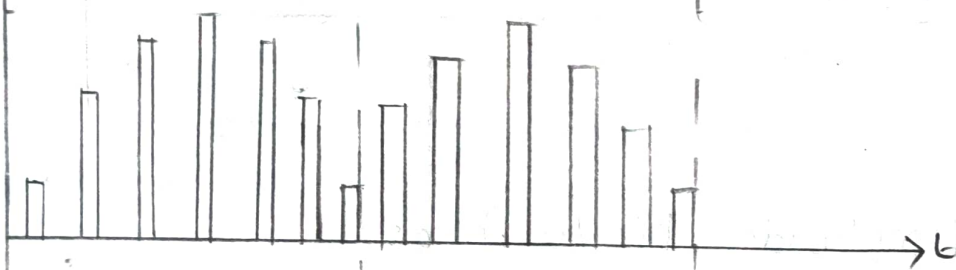
Frequency



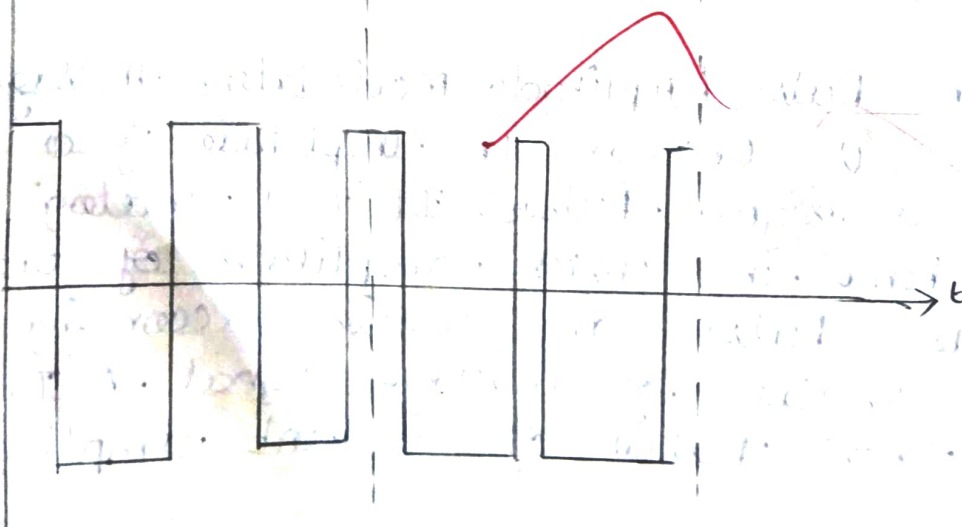
Message signal



Carrier signal



PAM



PWM

of these pulse is made to vary in accordance with that of modulating signal the result is that of multiplexing the train by modulating signal of these pulse highest corresponding to modulating wave.

The demodulated of the PAM waves signal is passed through a low pass filter having a cut-off frequency equal to highest frequency in modulating signal at output of filter is available the modulating signal along with DC component PAM has same signal to noise ratio

Pulse Position Modulation:

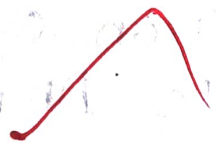
In Pulse Position modulation we have a fixed amplitude and pulse width of each other pulse, but position of each pulse is made proportional to amplitude of modulating signal at the instant.

~~In PPM~~ In PPM trainer board, Pulse Position modulation is generated by two monostable multivibrator. In this first multivibrator generator pulse width output and second generator pulse position modulation

frequency

PPM

DEMODULATED
SIGNAL



Initially synchronous clock from trainer given to pin-2 of 555 IC. Now we observe output at pin-3 of same 555 IC.

The position of each pulse is varied according to the already generated pulse width modulation. But PWM depends on the input AF signal. Therefore, on the amplitude of AF signal, position of each pulse is varied.

But width of each pulse remains constant because time constant of second monostable multivibrator is constant (fixed).

Pulse Width Modulation:

The pulse width modulation, we have fixed amplitude and starting time of each pulse, but the width of each pulse is made proportional to amplitude of modulating signal at instance.

In PAW trainer board, the pulse width modulation is generated by a monostable multivibrator using 555 IC connected in monostable mode. Initially synchronous clock from synchronous clock generator is given to

Pin-5. If we observe output at pin-3. The width of each pulse is varied in accordance with amplitude of AF signal.

The Demodulation of the pulse width modulation signal is simple process. Pulse an integrating circuit from which modulating signal is recovered, whose amplitude at any time is proportional to width of pulse. An amplifier circuit is used to amplify the recovered AF signal.

PROCEDURE:

- * Connect signal source 1KHz to A
- * Select sampling frequency to 8KHz
- * Connect Oscilloscope with input analog A₁ and with PAM modulator output E₁
- * For PAM demodulation connect signal source 1KHz (A₁) to A₁ PAM Output (E₁) to Receiver input (I₁) and PAM demodulation output (B₁) to LPF input B
- * For PWM modulation connect Oscilloscope with input analog signal A₁ and with generator
- * For PWM Demodulation connect the source 1KHz (A₁) to A. PWM output (E₁) to Receiver output (E) and PWM demodulator output (B₁) to LPF signal input B

* For PPM, Repeat Procedure of PWM Modulation to derive Proper PPM.

* For demodulation is connect signal source 1 kHz (A) to A. PPM Output (E₂) to receiver Input (E) and PPM demodulator output (B₃) to LPF Input (B).

* Connect the Oscilloscope with LPF Output E₁.

5/25/25
RESULT:

Thus the output waveform for PAM, PPM and PWM Modulation and Demodulation was Verified.

Implement Signal Sampling And Reconstruction

Aim:

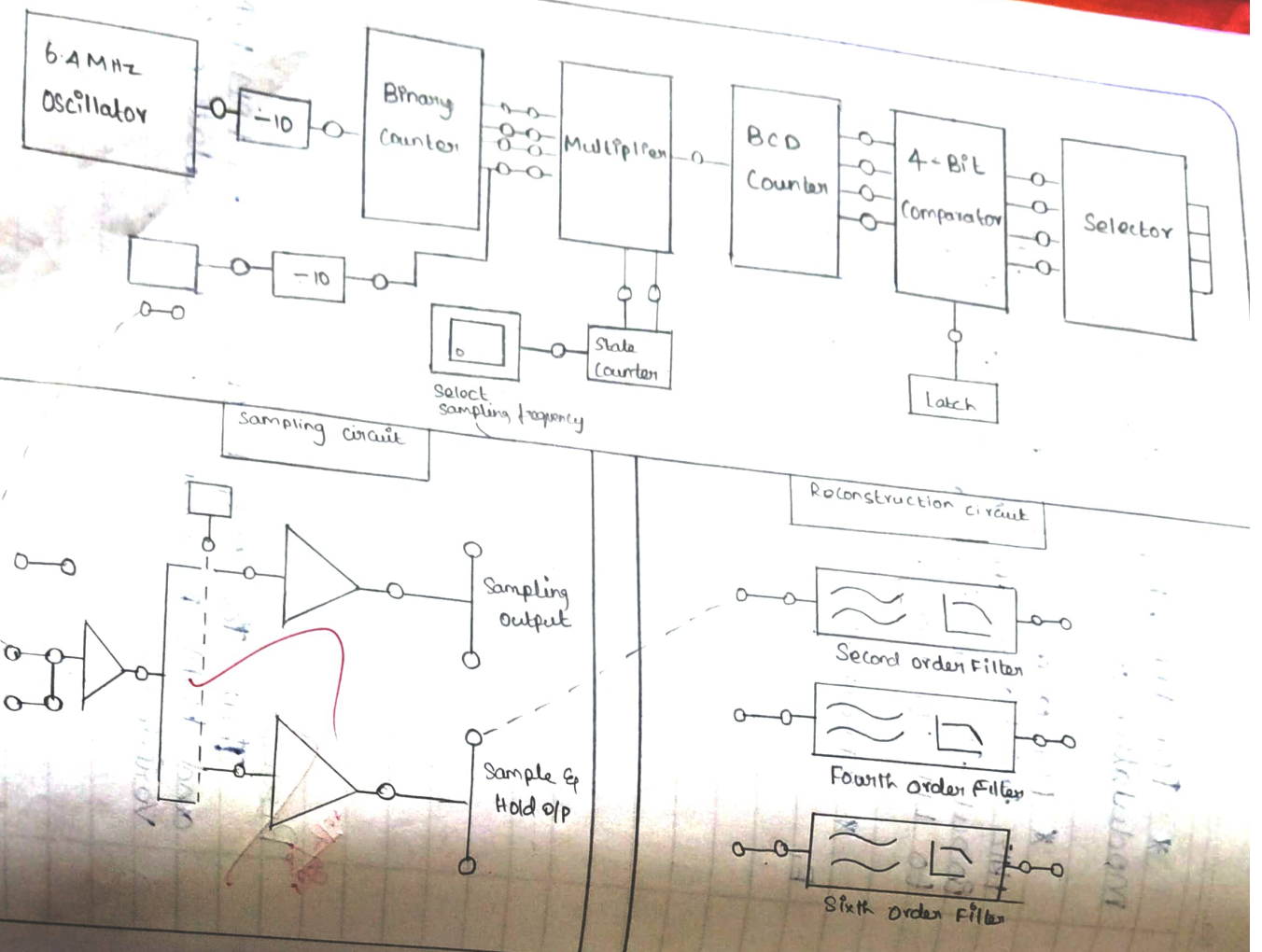
To analyse Process of Sampling and reconstruction using Trainer Kit

Apparatus Required:

S.NO	Apparatus	Range	Quantity
1.	Sampling and reconstruction kit		1
2.	DSO	(0-30)MHz	1
3.	Power cord		
4.	Patch cord		

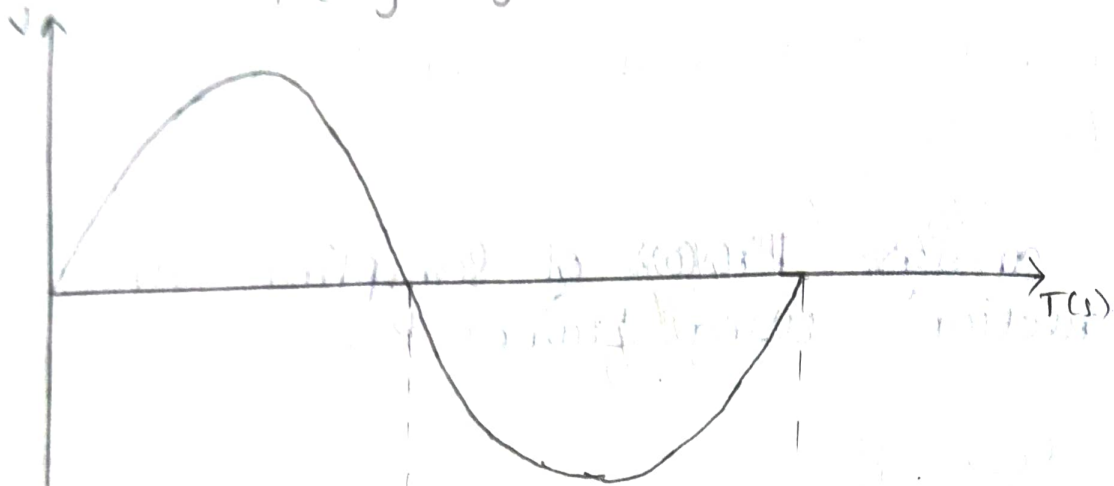
Theory:

Sampling is process of conversion of analog signal to discrete signal. Sampling shows that a continuous-time band limited signal may be represented perfectly by its samples at uniform intervals of T seconds. If this small enough. In other words, the continuous time signal may be reconstructed perfectly from its sample.

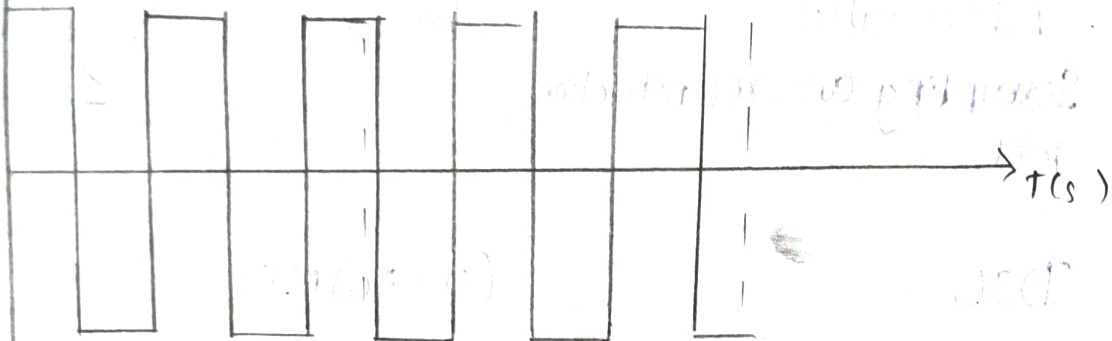


MODEL GRAPH:

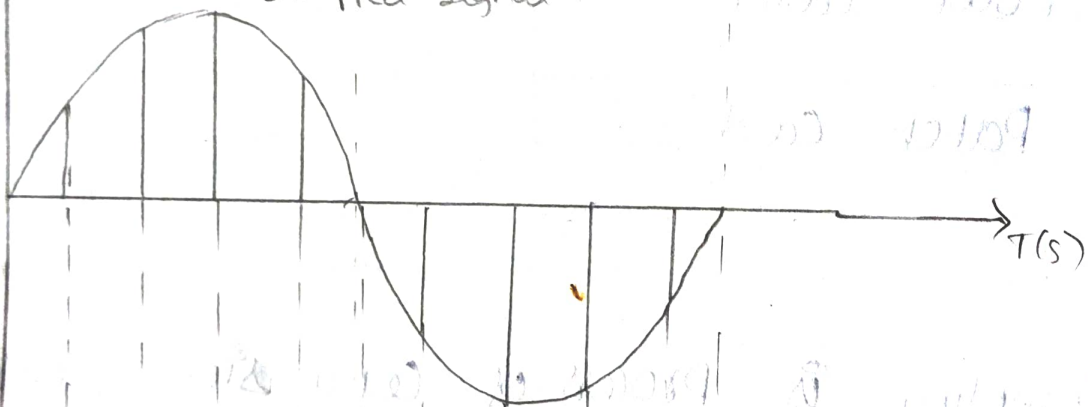
Message signal



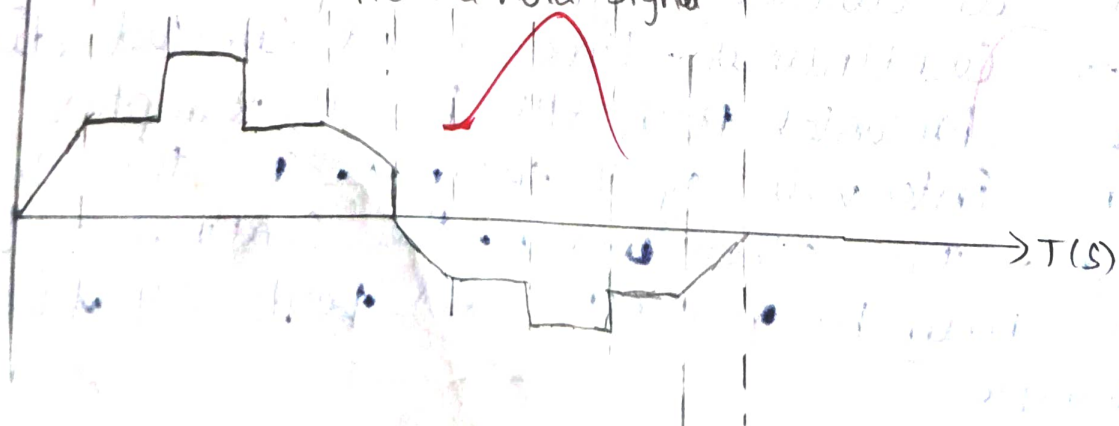
Carrier signal



Sampled signal



Sample and hold signal



Sampling at a high enough rate is information lossless

Sampling Theorem:

It states that the band limited signal of finite energy which has no frequency component higher than ' ω ' Hertz is completely described by specified the values of signal at instant of time separated.

$$f_s > 2f_m$$

$$g_s(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \cdot \delta(t - nT_s)$$

$g_s(t)$ = The Ideal Sampled Signal.

$$f_s = \frac{1}{T_s} ; \text{Sampling rate}$$

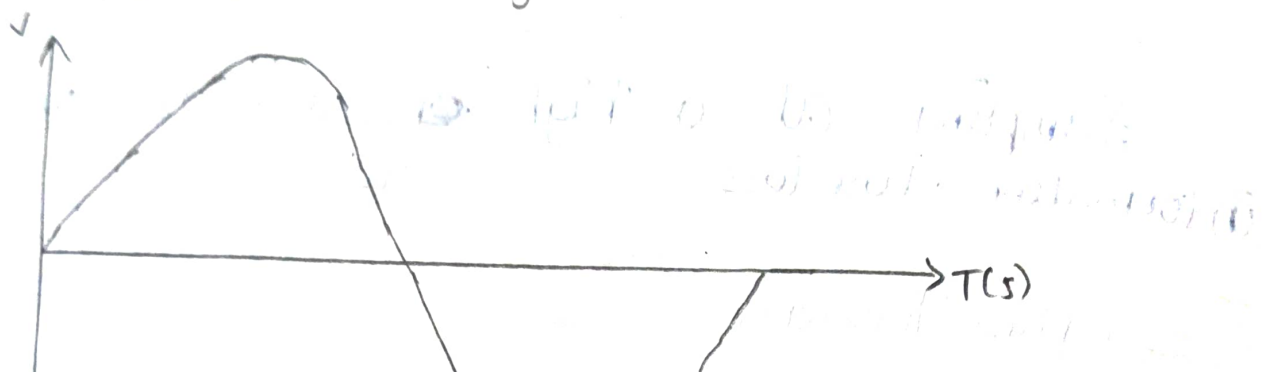
$$T_s = \text{Sampling period}$$

Procedure:

* Connections are given as per the block diagram.

* Take Sine wave as input of 1 KHz from signal generator block.

Reconstructed signal.



The reconstructed signal is a periodic waveform that starts at the origin (0,0), rises to a peak, crosses the horizontal axis, reaches a trough, and then returns to the axis. The waveform is smooth and continuous.

$$f(t) = \sin(2\pi t)$$

$$f(t) = \sin(2\pi t)$$

$$f(t) = \sin(2\pi t)$$

$$f(t) = \sin(2\pi t)$$

$$f(t) = \sin(2\pi t)$$

$$f(t) = \sin(2\pi t)$$

$$f(t) = \sin(2\pi t)$$

$$f(t) = \sin(2\pi t)$$

* Observe the carrier wave form and note down the amplitude and time period of the signal

* Observe the sampled signal and note down the amplitude and time period of the signal.

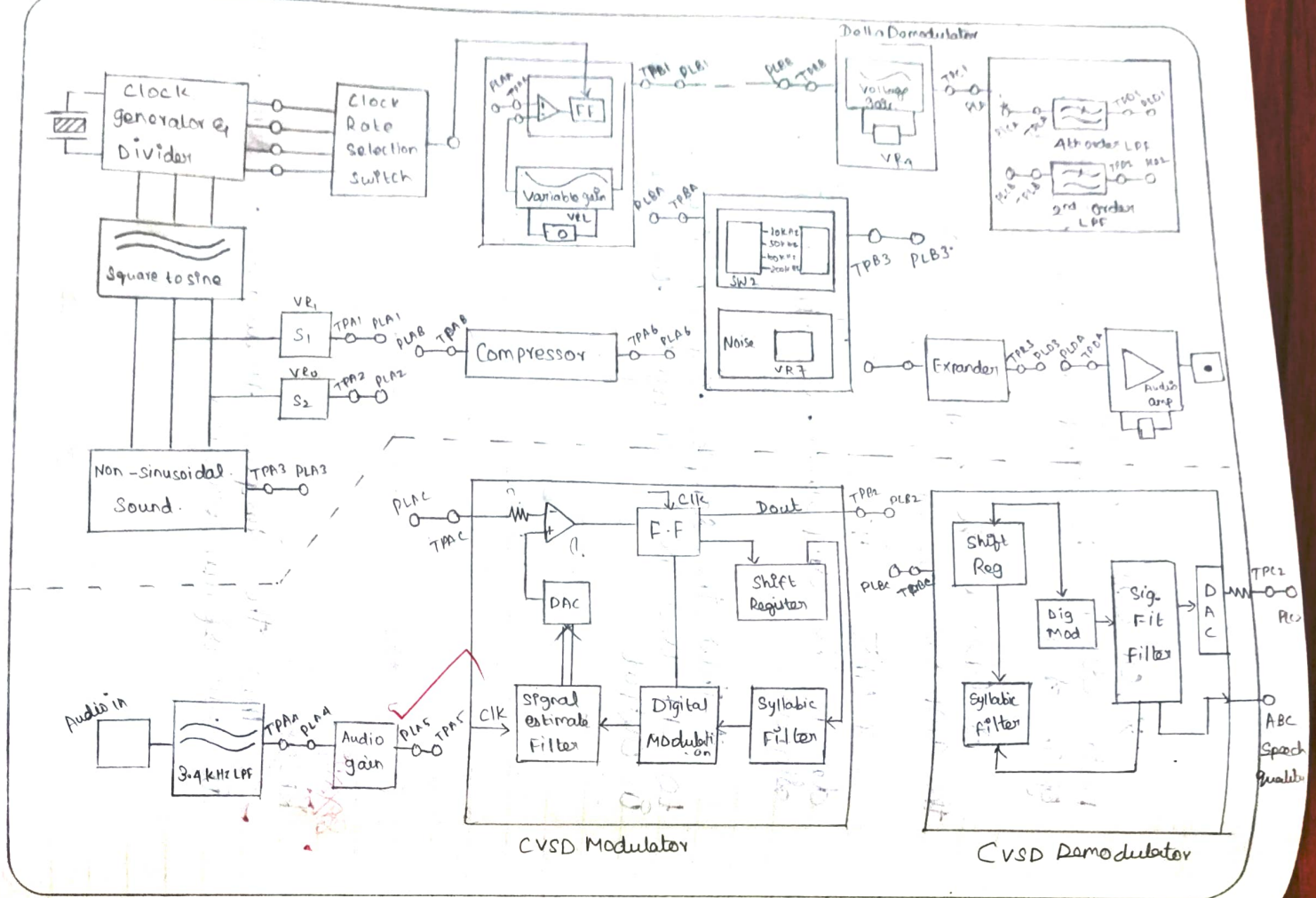
* Observe the sampled and add signal and down the amplitude and time period of the signal

* Thus the sampled signal is given an input to LPF and the reconstructed wave form obtained in output.

~~8/5/25~~
RESULT:

Thus the sampling modulation and demodulation is successfully executed.

BLOCK DIAGRAM:



Expt. No. 3

Date 19
Page No. 19

IMPLEMENT DELTA MODULATION AND DEMODULATION

Aim:

To study the operation of delta modulation and demodulation

Apparatus Required:

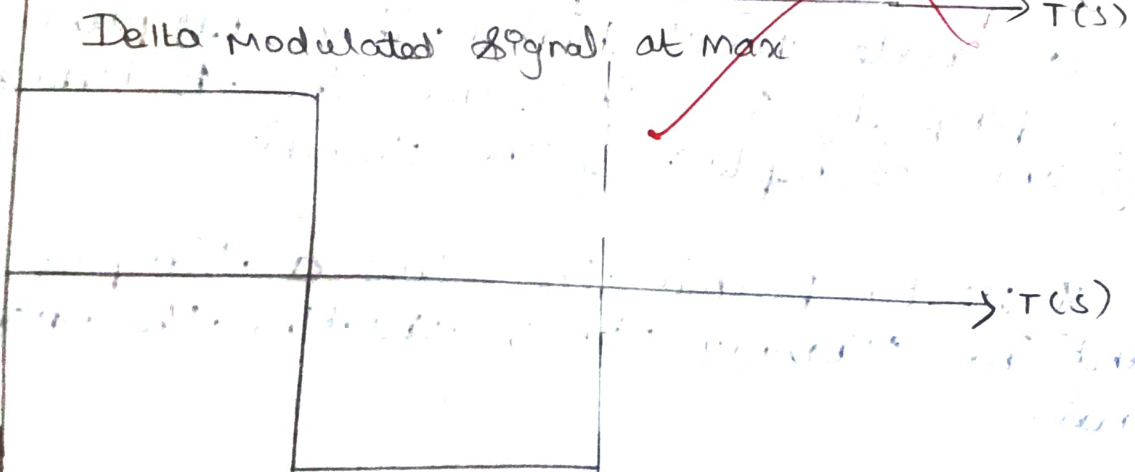
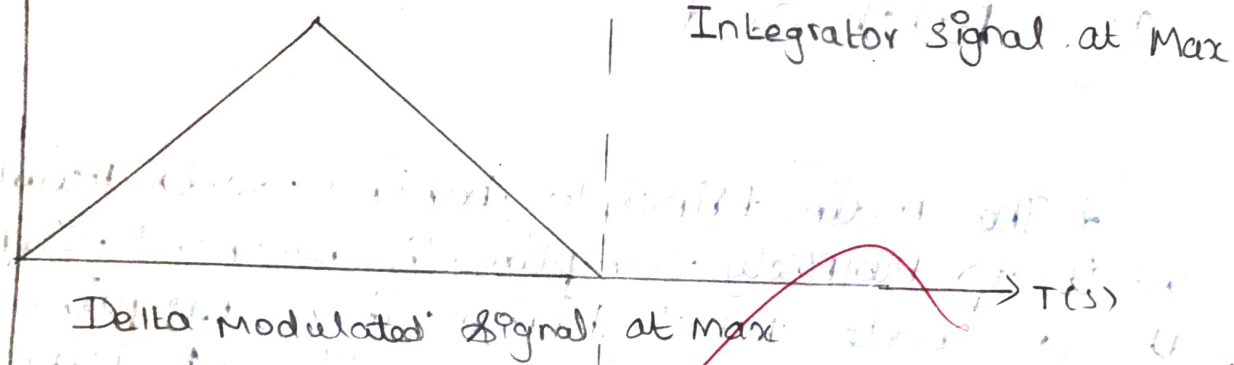
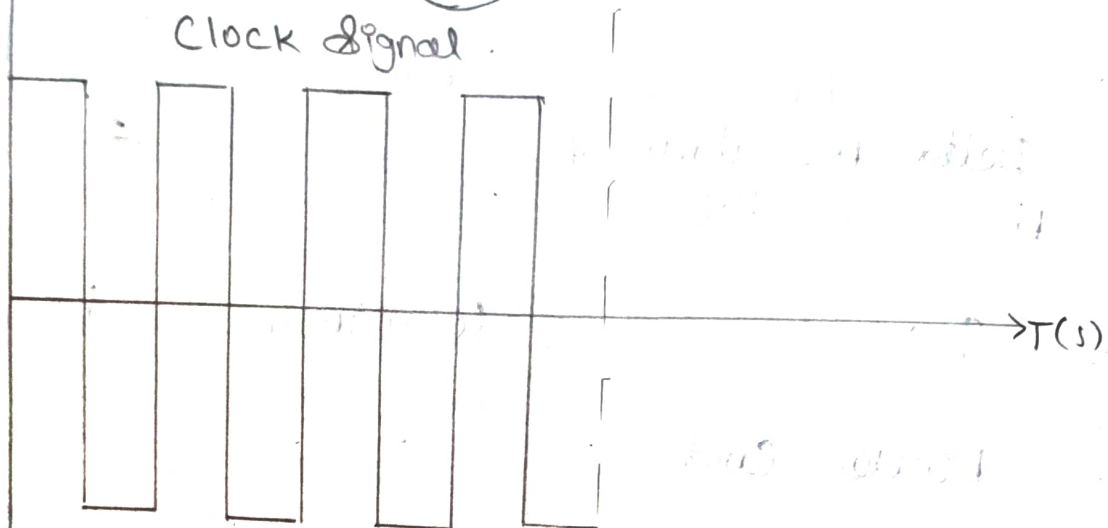
S. NO	Apparatus	Range	Quantity
1.	Delta modulation & Demodulation kit		1
2.	DSO	(0-30)MHz	1
3.	Power cord	-	-
4.	Patch cord	-	-

Theory:

* The main principle behind delta modulation and its to purposely sampling the baseband signal at a rate much higher than Nyquist rate properly to increase the between adjacent samples of the signal.

* As to present the use a simple quantizing strategy for constructing the encoded signal.

MODEL GRAPH:



* Delta modulation is a process of converting analog signal into one bit codes. In DM only one bit is sent per sample. This bit indicates whether the signal is larger or smaller than the previous sample.

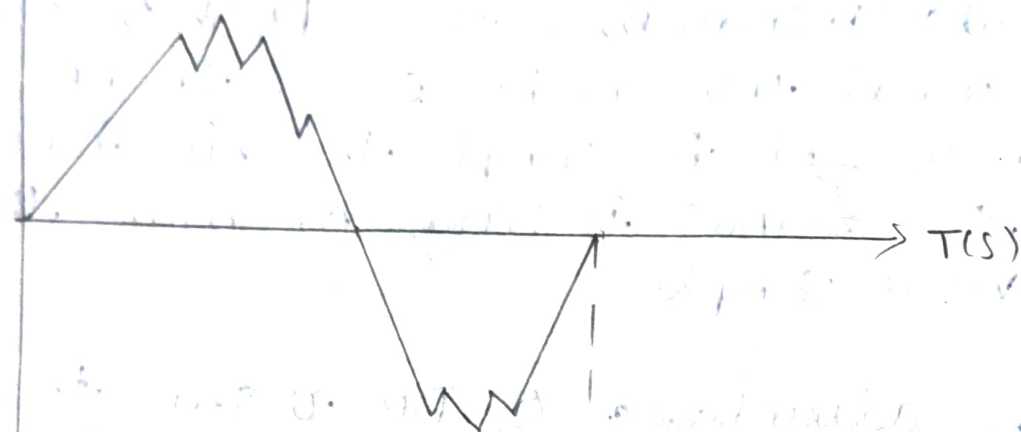
* The advantages of DM is that the modulator and demodulator circuits are much simpler than those used in traditional PCM signal.

* The purpose behind this form of modulation is to minimize the effect of noise without increasing the number of signal-to-noise ratio improving system performance.

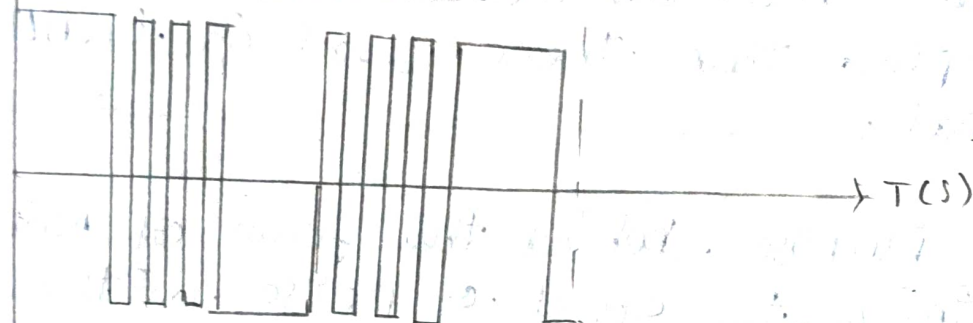
* The idea behind DM is to take samples close enough to each other so that each sample amplitude does not vary by more than signal step size.

* Then instead of sending a binary code representing the step size a signal sample size has increased or decreased by a single step.

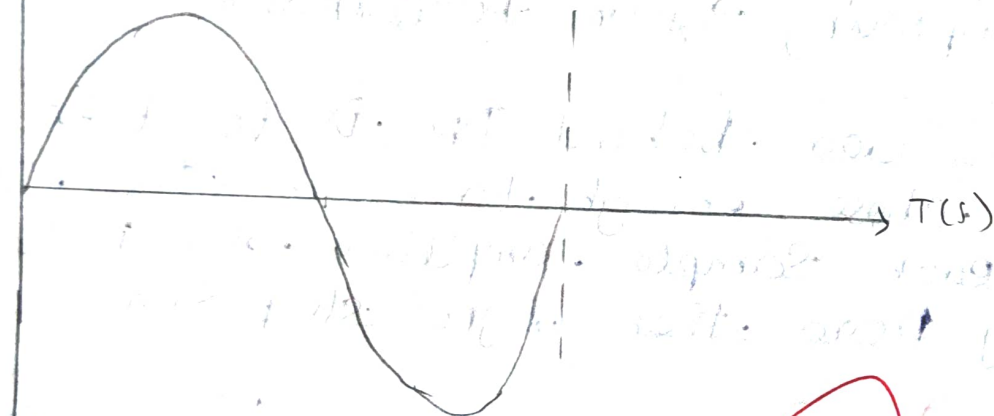
Integrator at Min



Delta modulated signal at Min



Demodulated signal



PROCEDURE:

* Connect the AC adapter to the main and the other side to the experiment trainer kit

* Switch ON experimental board

* Connecting the modulating signal to Input by DM and observe the same on channel of a dual trace oscilloscope.

* Connect clock signal to the DM circuit

* Observe the DM output to the Delta modulation

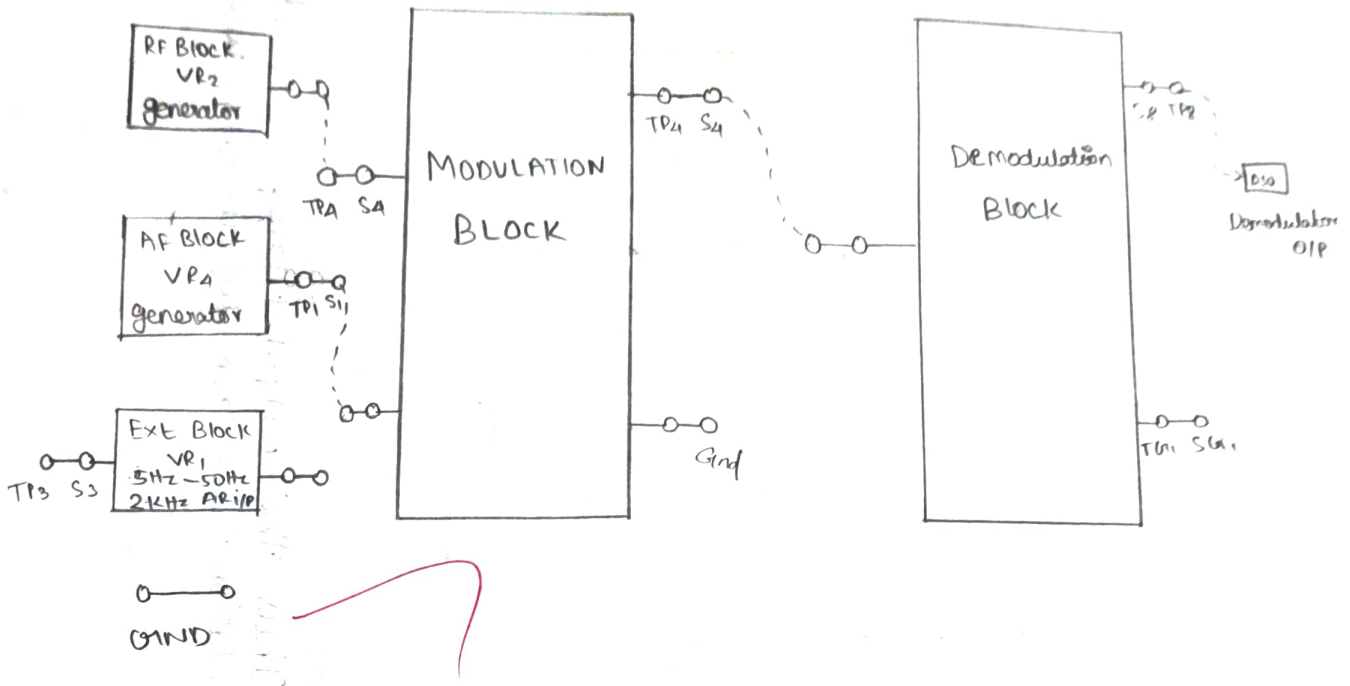
* Connect DSO output to the channel

* Observe the demodulating output with and without PC filter or CRO

~~20/7/25~~
22/7/25
RESULT:

Thus the output waveform of delta modulation and demodulation was verified

BLOCK DIAGRAM:



Expt. No. 4

Date: 24
Page No. 24

IMPLEMENT AM - MODULATOR AND DEMODULATOR

Aim:

To study the function of amplitude modulation and demodulation (under, over, critical modulation)

Apparatus Required:

S.No	Apparatus	Range	Quantity
1.	AM trainer kit	-	1
2.	DSC	(0-30) MHz	1
3.	Power cord	-	1
4.	Connecting wire	-	few
5.	Patch cord	-	few

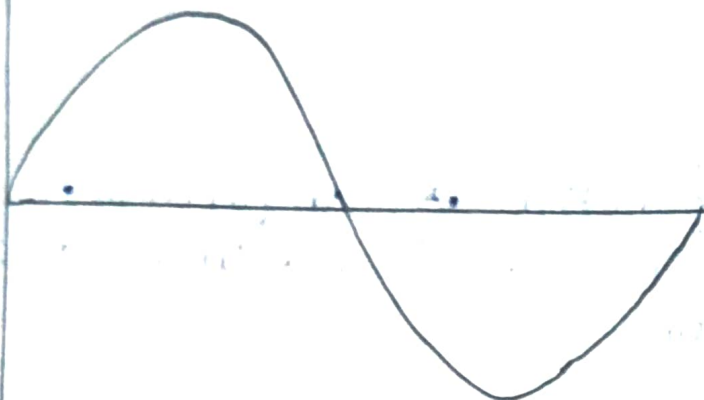
Theory:

Modulation:

Modulation is defined as process of changing the characteristic of the carrier signal in accordance with intensity of the message signal.

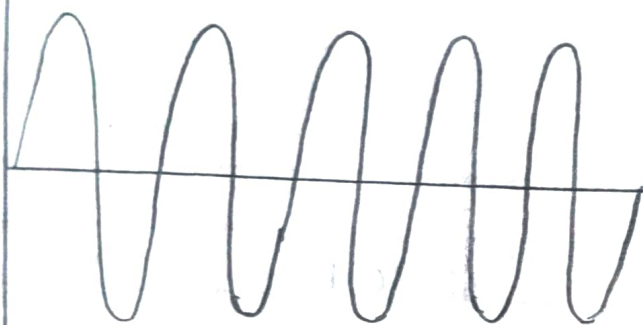
MODEL GRAPH

V

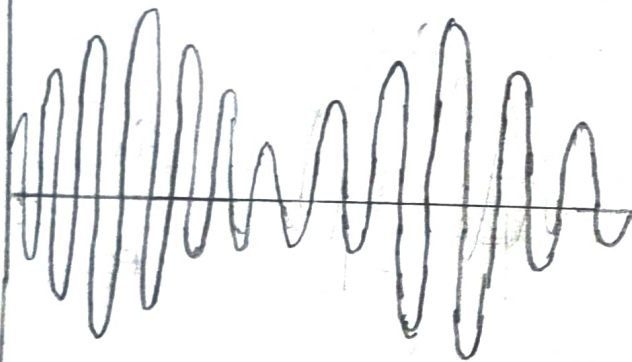


Message signal

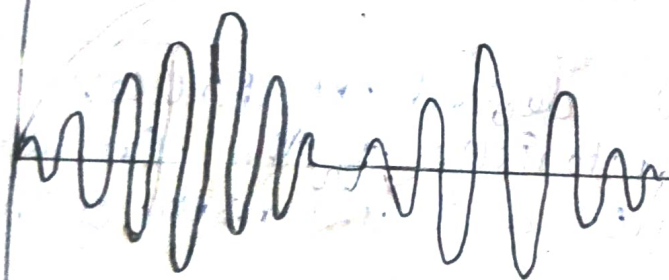
Carrier signal



Under modulated signal



Critically modulated signal



Amplitude modulation is defined as a system of modulation in which the amplitude of the carrier is varied in accordance with amplitude of the message signal.

De modulation is the reverse process to recover the original message signal at the receiver.

AM DEMODULATION:

AM demodulation is the process of extracting the original message signal from an Amplitude Modulated Carrier wave.

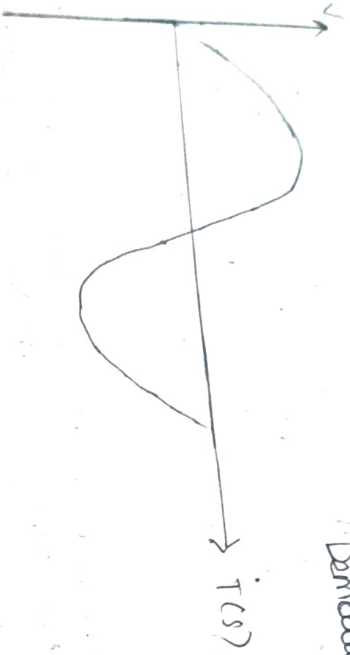
Frequency and Phase of the carrier signal remain constant.

$$m(t) = A_m \cos \omega_m t$$

$$c(t) = A_c \cos \omega_c t$$

$$x_c(t) = (A_c + A_m \cos \omega_m t) \cos \omega_c t$$

Demodulated signal



Expt. No.

Date
Page No. 31

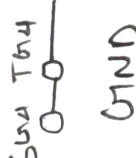
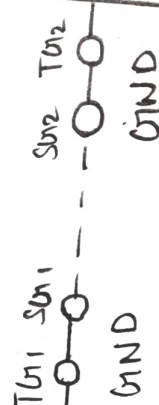
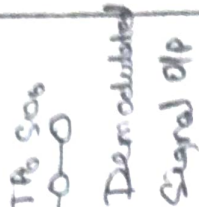
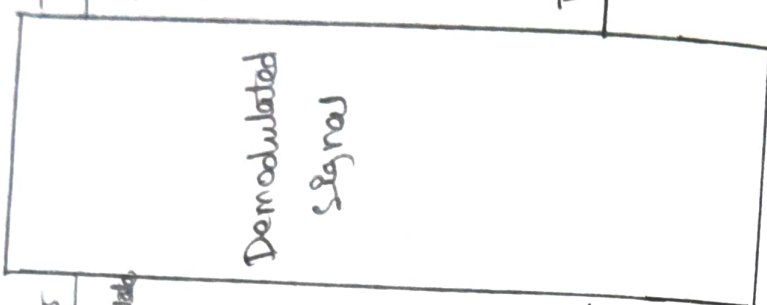
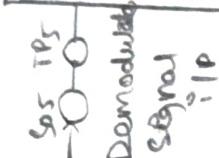
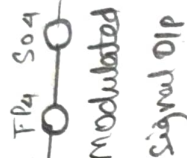
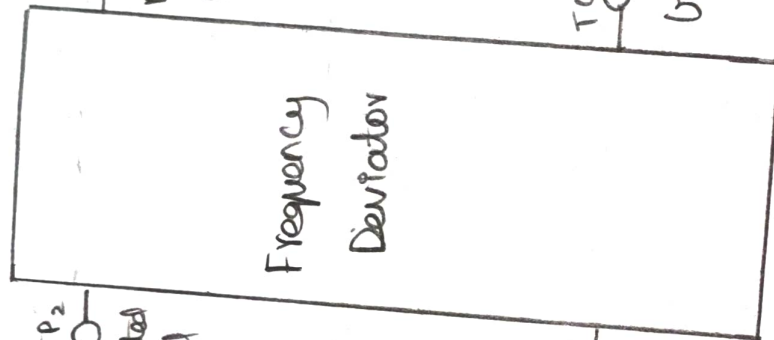
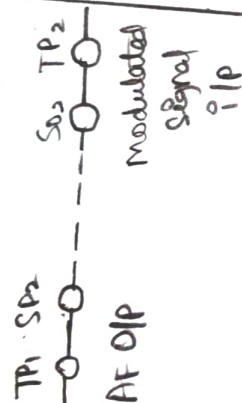
PROCEDURE:

- * Switch on the trainer kit and check the output of the carrier generator on oscilloscope.
- * Connect ground 1KHz with 2 volts AF signal at AF input to the modulator circuit.
- * Connect the carrier signal at the input of the modulator circuit.
- * Observe the modulated output signal at an output.
- * Vary the modulating frequency and amplitude and observe the effect on the modulated wave form.
- * The depth of modulation can be varied using variable knob provided at AF input.
- * The Percentage modulation can be calculated using formula.

~~100~~ RESULT:

Thus the implementation of Am modulation and Demodulation is performed and the output is verified.

FM MODULATION AND DEMODULATION



IMPLEMENT FM MODULATION AND DEMODULATION

Aim:

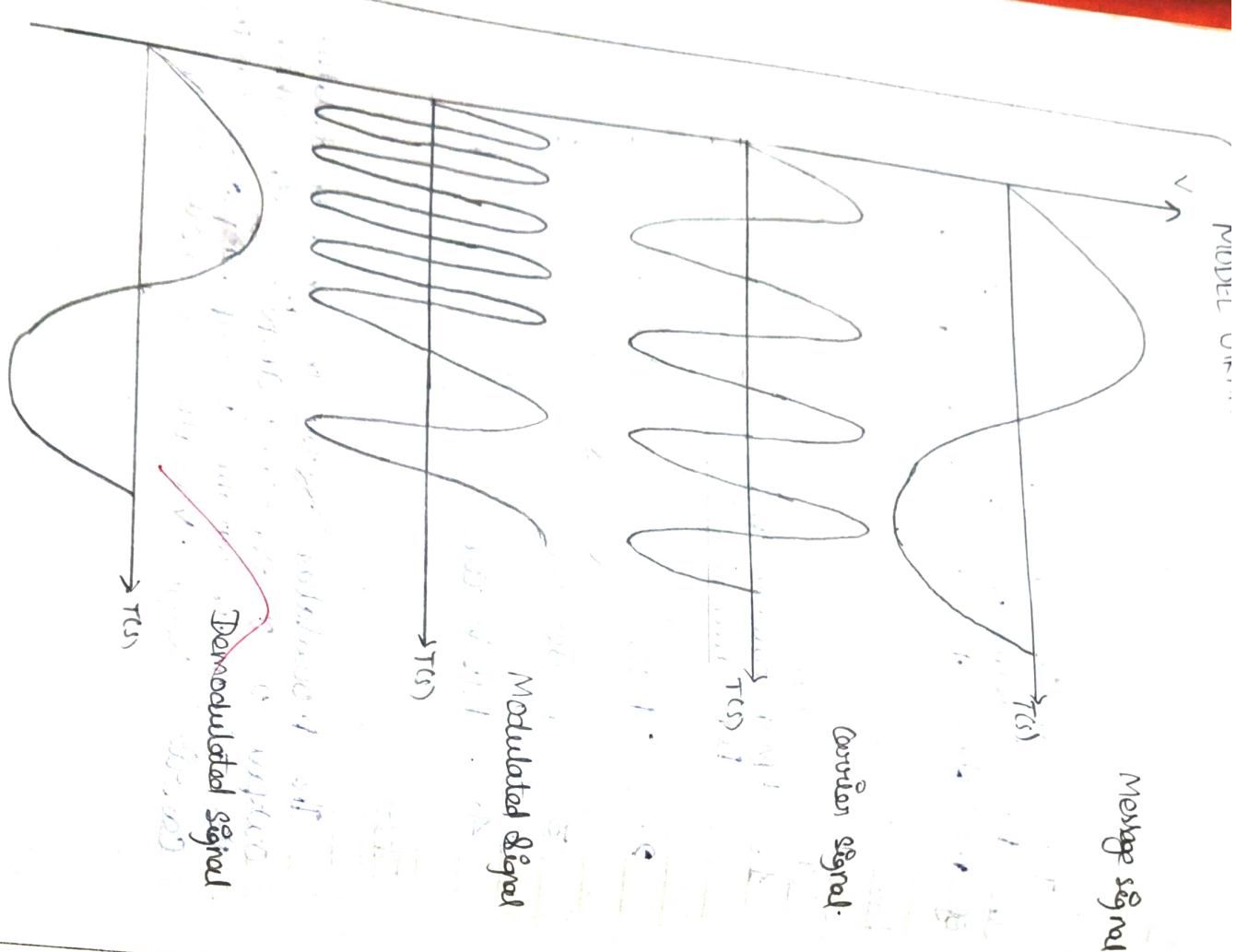
TO Study the Process of Frequency Modulation and Demodulation and to calculate the depth of modulation by varying the modulating voltage.

Apparatus Required.

S.NO	Apparatus	Range	Quantity
1.	FM modulation and Demodulation kit	-	1
2.	DSO	(0-30)MHz	1
3.	Power Cord	-	1
4.	Patch cord	-	few

Theory

The Modulation System in which modulation output is of constant amplitude in which the ~~signal~~ information is superimposed on the carrier through variation of carrier frequency.



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Date
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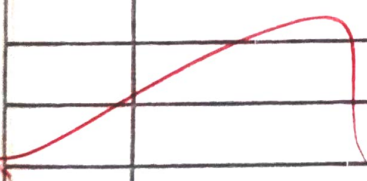
The frequency modulation is a non linear modulation process. Each spectral component of the base band signal gives rise to one or two spectral components are spectral from carrier by a signal frequency difference equal to the frequency of base band component. Mostly, the nature of the modulator is such that the spectral component which produces on the carrier frequency and base band frequency spectral component in modulated waveform depends on the amplitude.

The message signal $x_m(t)$ and sinusoidal carrier signal $x_c(t) = A_c \cos(2\pi f_c t)$ where f_c is the carrier frequency and A_c is carrier amplitude. The modulation combines the carrier with base band data signal to get transmitted signal

$$y(t) = A_c \cos(2\pi f_c t + 2\pi f_n \int x_m(t) dt)$$

where $f_n \neq k f_c$, A_m , k being the sensitivity of frequency modulator and A_m being amplitude of the modulating signal

PROCEDURE:

- *) Switch the experimental board or the trainer kit
 - *) Observe the FM modulator output without any modulator input which is carrier signal.
 - *) Connect modulating signal to FM modulator input and observe modulating signal and FM output on two channels of CRO or DSO
 - *) Adjust the Amplitude of Modulation signal until we get less distorted FM output
 - *) Apply FM output to FM demodulator and gain adjust the potentiometer in demodulator until the demodulator output is obtained
- 

Expt. No.....

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~~8X~~ RESULT:

Thus the FM Modulation and Demodulation was Implemented and the output waveform was successfully verified.

Function Generator

Signal
S₀
I/P
T₀

Pre-emphasis

Signal
T₁
O/P
S₁

Signal
S₂
I/P
T₂
Sum D₂

De-emphasis

Signal
T₃
O/P
Sum D₃

Function Generator

Signal
S₀
I/P
T₀

Pre-emphasis

Signal
T₁
O/P
S₁

DSO

Signal
S₂
I/P
T₂
Sum D₂

De-emphasis

Signal
T₃
O/P
Sum D₃

Expt. No. 6 ... Date ... Page No. ... 43

Pre-Emphasis And De-Emphasis

Aim:

To Perform Pre-emphasis and de-emphasis using the trainer kit

Apparatus Required:

S.No	Apparatus	Range	Quantity
1.	Trainer Kit	-	1
2.	Function generator	(0-30)MHz	1
3.	DSO	(0-30)MHz	1
4.	Power Cord	-	1
5.	Patch cord	-	1

Theory:

Pre-emphasis:

Pre-emphasis refers to boosting the relative amplitude of modulating voltage for higher audio frequency from 2 to approximately 15 kHz.

At the transmitter modulating is passed through a simple network called

PRE-EMPHASIS:

$$V_{in} = V_{pp} = 5V$$

S.No	Frequency (Hz)	Voltage (Vo)	$\frac{V_o}{V_{in}}$	$20 \log_{10} \left(\frac{V_o}{V_{in}} \right)$
1.	100	6.8	2.72	7.12
2.	200	6.8	2.72	7.12
3.	300	6.8	2.72	7.12
4.	400	6.8	2.72	7.12
5.	500	7.4	2.96	9.4
6.	600	8.56	3.45	10.7
7.	700	8.72	3.48	10.8
8.	800	9.04	3.61	11.1
9.	900	9.36	3.74	11.4
10.	1K	10.1	4.04	12.1
11.	2K	10.7	4.28	12.6
12.	3K	11.1	4.44	12.9
13.	4K	11.3	4.52	13.1
14.	5K	12.4	4.96	13.9
15.	6K	13.0	5.2	14.3
16.	7K	13.8	5.52	14.8
17.	10K	13.8	5.52	14.8
18.	20K	13.8	5.52	14.8
19.	30K	13.8	5.52	14.8
20.	40K	13.8	5.52	14.8

Pre emphasis

frequency

100Hz

1K

10K

100K

SEMI-LOG PAPER (CYCLES X 1000)

scale

x axis - frequency

y axis 1cm = 5 units

gain

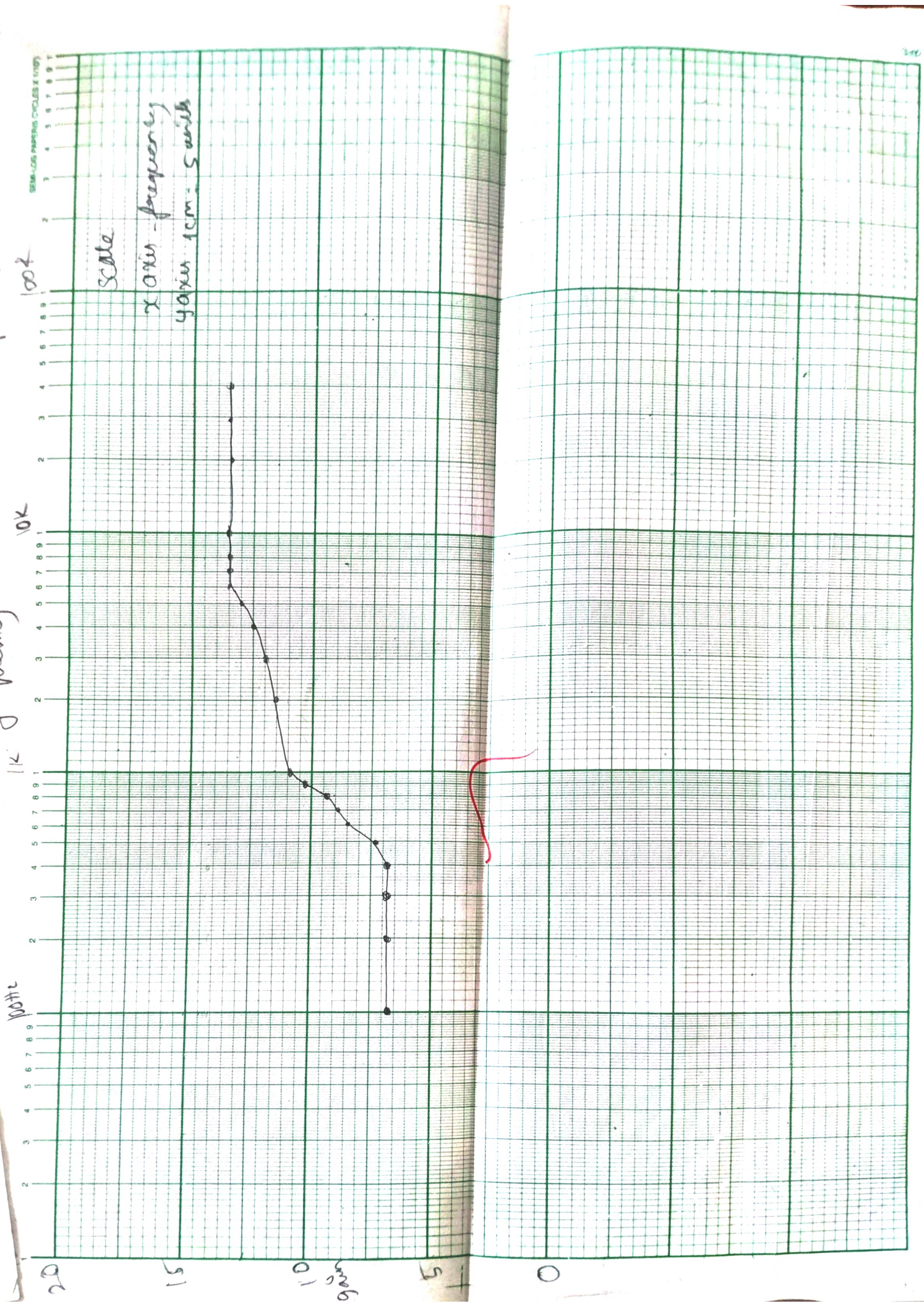
20

15

10

5

0



the higher frequency component than lower frequency components.

The simplest form of such a circuit is a simple high pass filter of the type A circuit has cutoff frequency of 2122 Hz . This means that frequencies higher than 2122 Hz . The amplitude with frequency at a 6 dB per octave.

De-emphasis:

De-emphasis means attenuating pre-emphasis signal by amount by which they are boosted. A cutoff frequency is attenuated at 6 dB per octave. As a result, pre-emphasis at transmitter is exactly offset by de-emphasis in the receiver, providing a normal frequency response. The combined effect of pre and de-emphasis is to increase high frequency components during transmission so that they will be stronger and not masked by noise.

DE-EMPHASIS

$V_{in} = 5V$

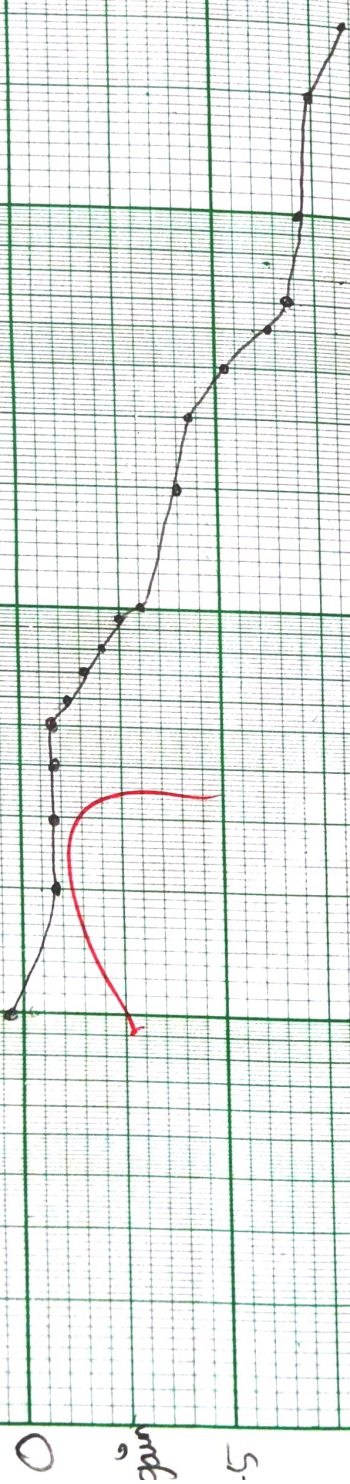
Sr No	FREQUENCY	VOLTAGE V_o	$\frac{V_o}{V_{in}}$	$20 \log_{10} \left(\frac{V_o}{V_{in}} \right)$
1	100	2.6	1.04	0.34
2	200	2.3	0.92	-0.72
3	300	2.3	0.92	-0.72
4	400	2.3	0.92	-0.72
5	500	2.3	0.92	-0.72
6	600	2.1	0.84	-1.51
7	700	2.0	0.84	-1.91
8	800	1.9	0.76	-2.3
9	900	1.82	0.72	-2.8
10	1K	1.7	0.68	-3.3
11	2K	1.59	0.63	-4.01
12	3K	1.53	0.61	-4.2
13	4K	1.4	0.56	-5.03
14	5K	1.2	0.48	-6.37
15	6K	1.15	0.46	-6.7
16	7K	1.08	0.432	-7.2
17	10K	1.02	0.42	-7.5
18	20K	1.02	0.42	-7.9

Deemphasis frequency

100 Hz 1K 10K 100K

Scale

x axis - frequency
y axis - cm = 5 units

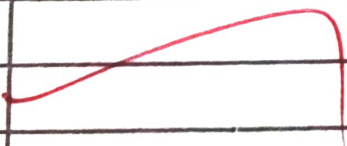


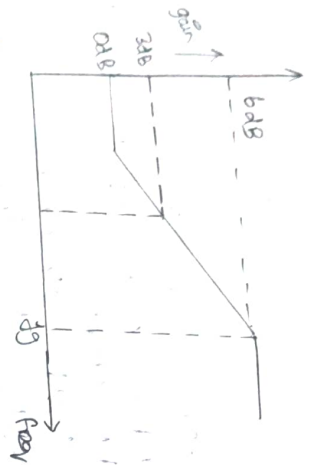
Procedure:Pre-emphasis:

- * Switch ON Trainer Kit
- * Give input signal from function generator to Pre-emphasis
- * Observe output using DSO
- * By varying the input frequency note down corresponding reading

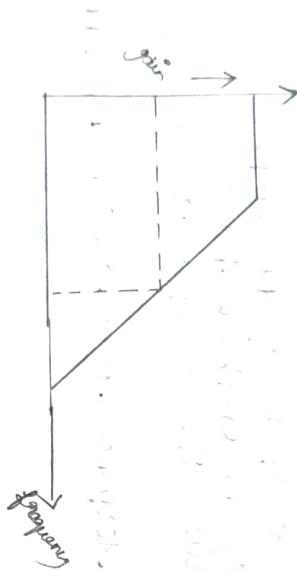
De-emphasis:

- * Give output of Pre-emphasis as a input of De-emphasis block
- * Observe the output of de-emphasis circuit
- * The response will be similar to LPF





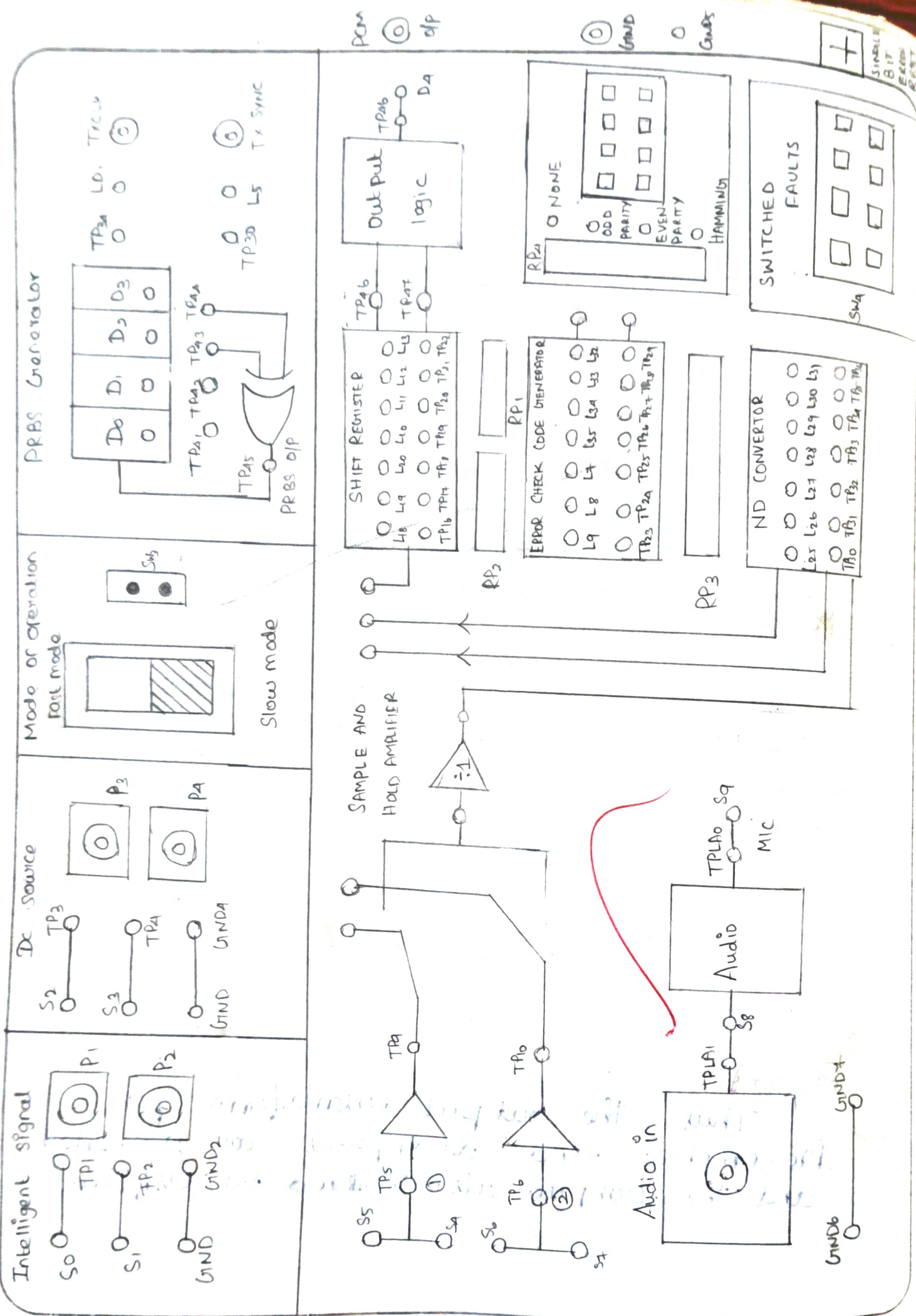
DE-EMPHASIS:



Result:

Thus the output waveform for pre-emphasis and de-emphasis are generated and the output also verified successfully.

BLOCK DIAGRAM:



Pulse Code Modulation and Demodulation

Aim:

To convert an analog signal into a pulse digital signal using PCM system and to convert the digital signal into analog signal using PCM demodulated system.

Apparatus Required

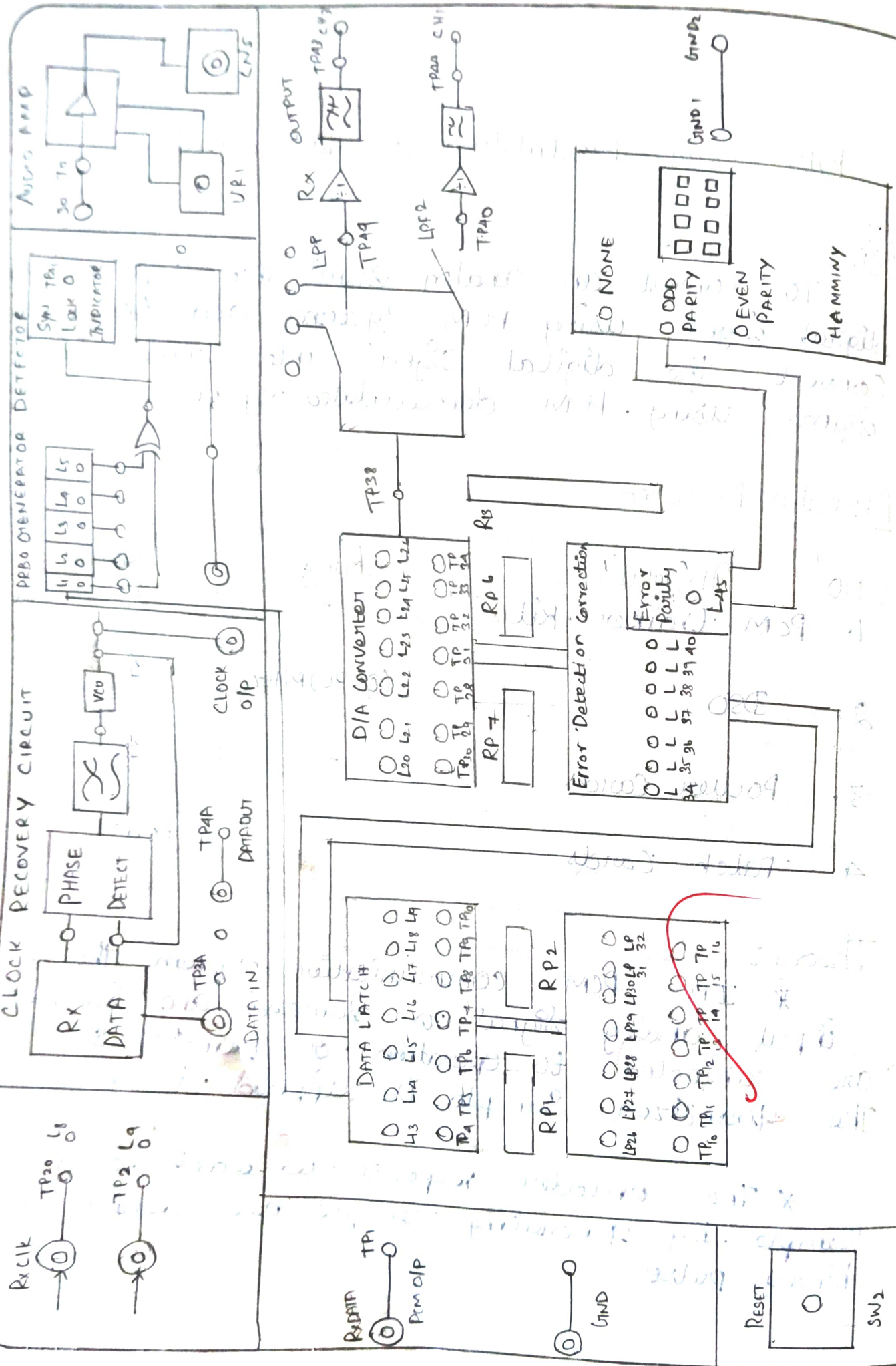
S.NO	Apparatus	Range	Quantity
1.	PCM trainer kit	-	1
2.	DSO	(0-30) MHz	1
3.	Power cords	-	1
4.	Patch Cords	-	few

Theory:

* In PCM communication system the input analog signal is sampled and samples are subjected to operation of quantizers. The quantizer sample is applied to encoder.

* The encoder response to each such sample by generating unique and identifiable binary pulse.

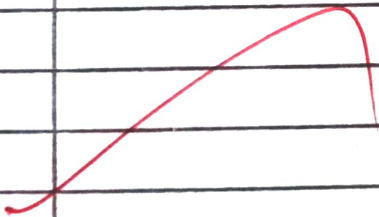
Block Diagram



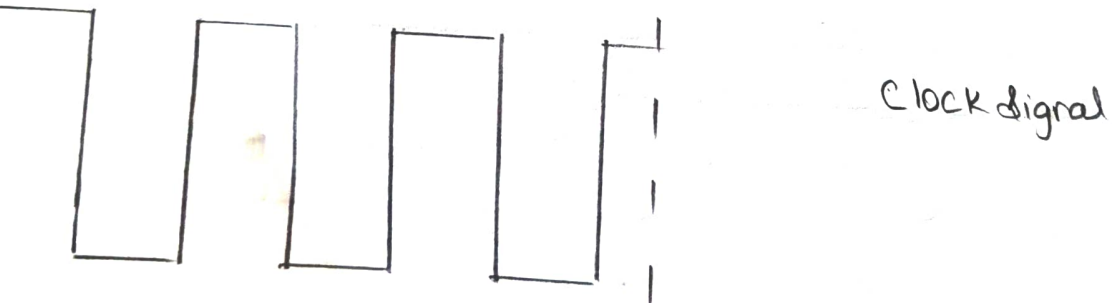
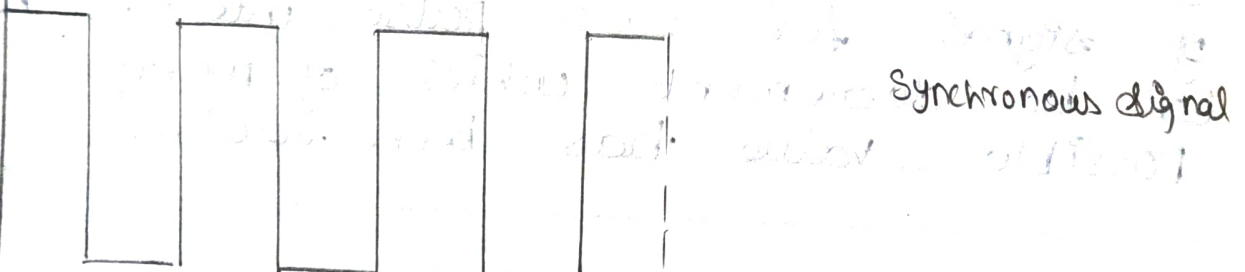
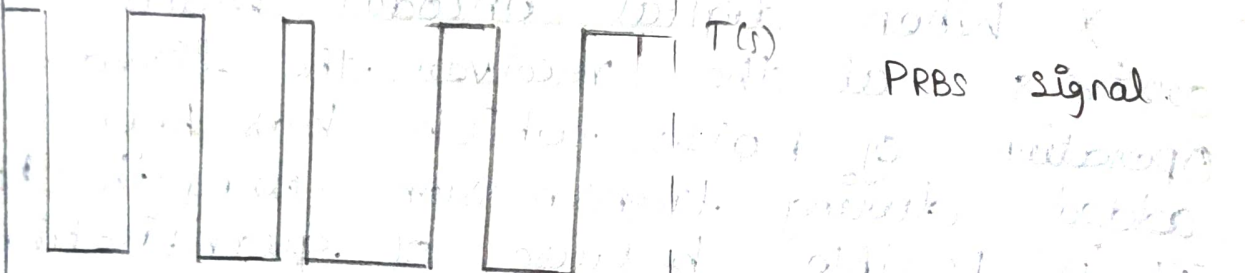
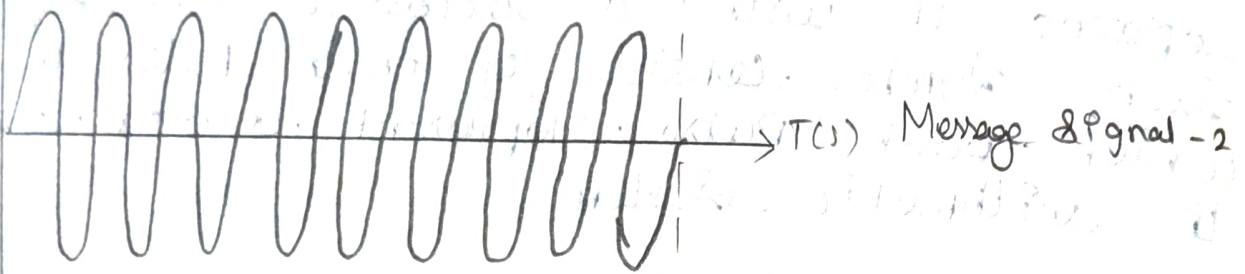
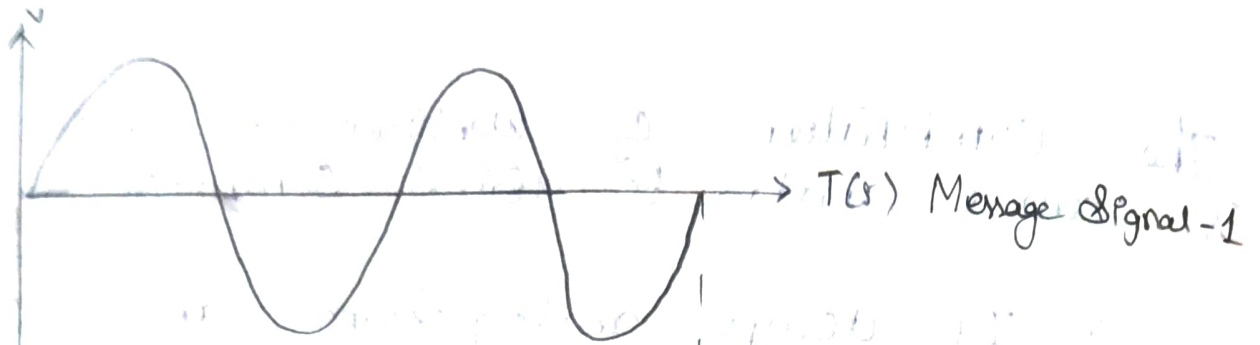
The combination of quantiser and encoder is called analog to digital converter.

* It accepts analog signal and replaces it with a successive code signal. Such a symbol consists of train pulse in which each pulse represents a digital in arithmetic system.

* When digital encoder signal arrives at the receiver, the fixed operation of noise which has been added during transmission along the channel. It is possible because of quantisation of signal for each pulse interval has to be determined which of many possible values has been received.



MODEL GRAPH:



Procedure:

* Connect the AC adapter to mains and other side to the experience trainer

* Power on the force tech Pulse Code modulation and demodulation system

* Measure the frequencies of the clock

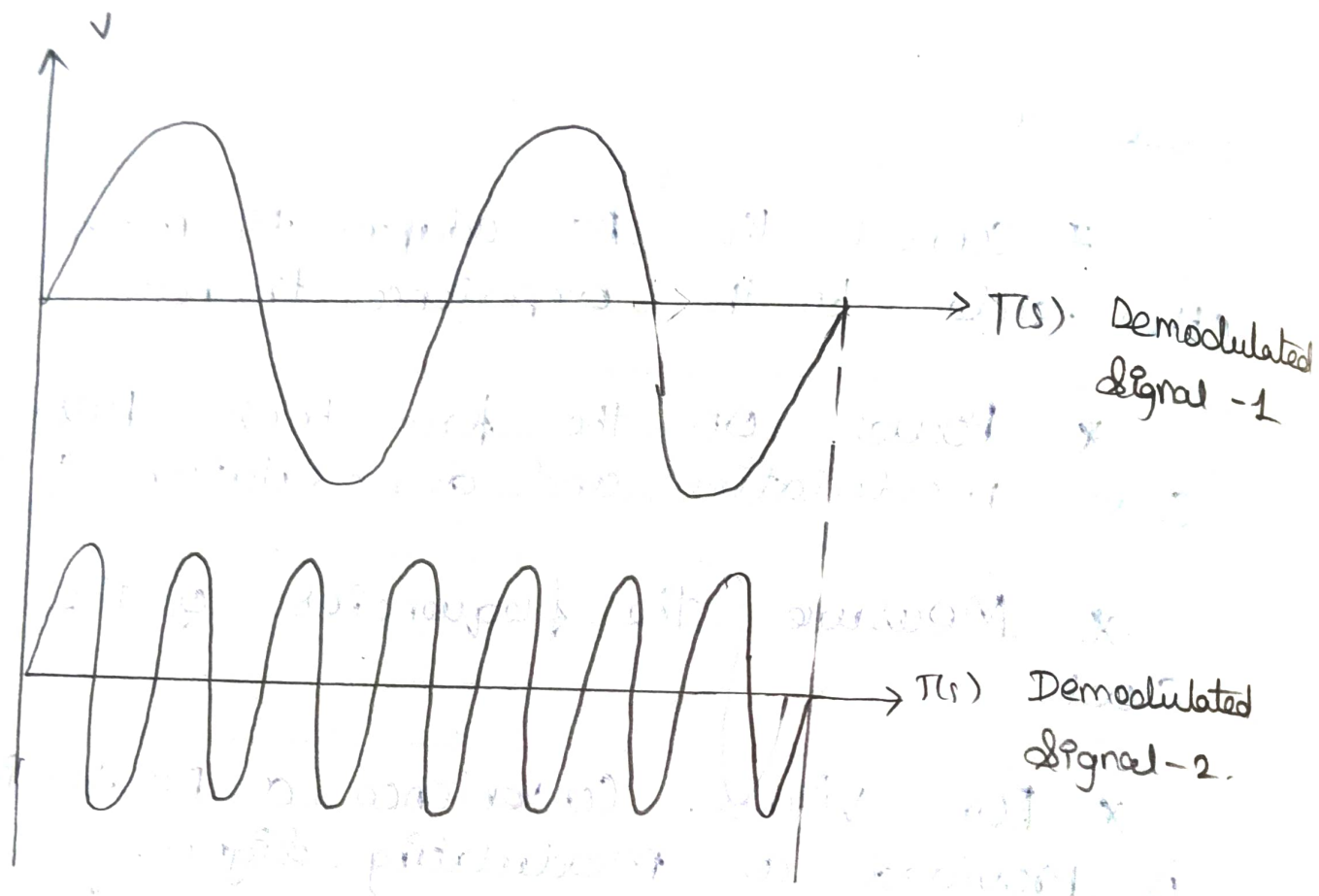
* For visual convenience a DC variable is provided as modulating signal.

* Connect the clock to the clock input signal.

* Connect the DC to input of AF input signal in modulator section.

* Note down the binary word from the LED's

* To see the data which is transmitted through channel connected the sampling clock to the channel of oscilloscope and serial data of channel



* Now apply the modulation signal or Input

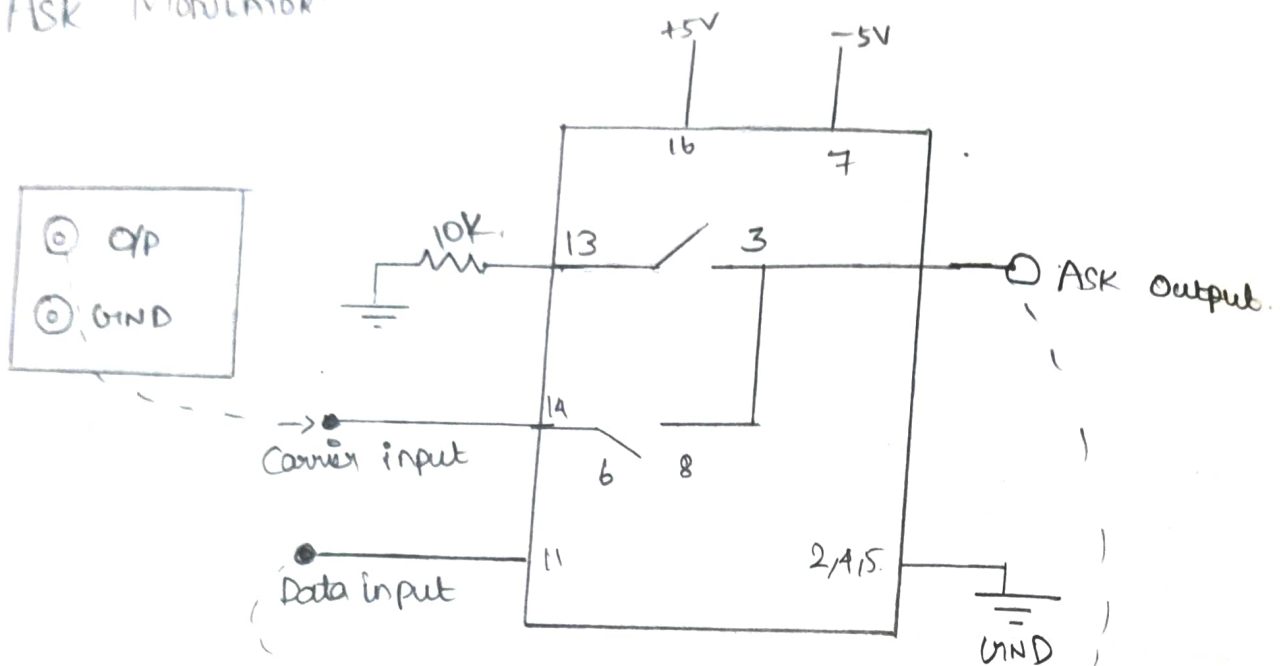
* Observe the waveform at output of DSO which is in quantised level. and observe the demodulated output

RESULT:

Thus the output waveform for pulse code modulating and demodulation was Verified.

BLOCK DIAGRAM:

ASK Modulator



ASK DEMODULATOR

